Problem Definition

This project analyzes student exam performance data to identify distinct groups of students based on their study habits, sleep patterns, attendance, and academic performance. The goal is to discover natural groupings in data that can help educators understand different student profiles and potentially tailor interventions or support strategies.

The dataset represents an educational environment where student success is influenced by multiple behavioral and academic factors. By applying clustering algorithms, we aim to segment students into meaningful groups that share similar characteristics, which could inform personalized learning approaches.

Data Description

The dataset contains 1000 student records with the following attributes:

Attribute	Type	Description
student_id	Categorical	Unique identifier
hours_studied	Continuous	Hours per week
sleep_hours	Continuous	Hours per night
attendance_percent	Continuous	Percentage (0-100)
previous_scores	Continuous	Historical performance
exam_score	Continuous	Current exam result

Data Preprocessing

Data Quality and Cleaning

The dataset was examined for quality issues:

- Missing values: No missing values were found in the dataset
- Outliers: Box plots were used to identify potential outliers in continuous attributes
- Duplicates: No duplicate records were identified

The data was found to be complete and consistent, requiring no cleaning operations.

	student_id	hours_studied	sleep_hours	attendance_percent	previous_scores	exam_score
0	S001	8.0	8.8	72.1	45	30.2
1	S002	1.3	8.6	60.7	55	25.0
2	S003	4.0	8.2	73.7	86	35.8
3	S004	3.5	4.8	95.1	66	34.0
4	S005	9.1	6.4	89.8	71	40.3
5	S006	8.4	5.1	58.5	75	35.7
6	S007	10.8	6.0	54.2	88	37.9
7	S008	2.0	4.3	75.8	55	18.3
8	S009	5.6	5.9	81.6	84	34.7
9	S010	1.3	8.9	66.8	70	24.7

(200, 6)

Figure 1: Image of Clean Data Set

```
Missing values: student_id
hours_studied         0
sleep_hours           0
attendance_percent    0
previous_scores       0
exam_score            0
dtype: int64

No duplicate Student IDs found

hours_studied:
Range: [1.00, 12.00]
IQR bounds: [-4.75, 17.25]
Outliers: 0 (0.0%)

sleep_hours:
Range: [4.00, 9.00]
IQR bounds: [1.21, 12.11]
Outliers: 0 (0.0%)

attendance_percent:
Range: [50.30, 100.00]
IQR bounds: [24.36, 125.26]
Outliers: 0 (0.0%)

previous_scores:
Range: [40.00, 95.00]
IQR bounds: [15.00, 119.00]
Outliers: 0 (0.0%)

exam_score:
Range: [17.10, 51.30]
IQR bounds: [15.62, 52.62]
Outliers: 0 (0.0%)
```

Figure 2: Data Cleaning Conclusion

Discretization

```
Exam Score Binning:
  Low: 0 - 30.70
  Medium: 30.70 - 36.77
  High: 36.77 - 100

Distribution of discretized exam_score:
exam_score_binned
Low       68
Medium    66
High      66
Name: count, dtype: int64

Sample of original vs discretized:
   exam_score exam_score_binned
0        30.2               Low
1        25.0               Low
2        35.8            Medium
3        34.0            Medium
4        40.3              High
5        35.7            Medium
6        37.9              High
7        18.3               Low
8        34.7            Medium
9        24.7               Low
```

Figure 3: Showing Discretization

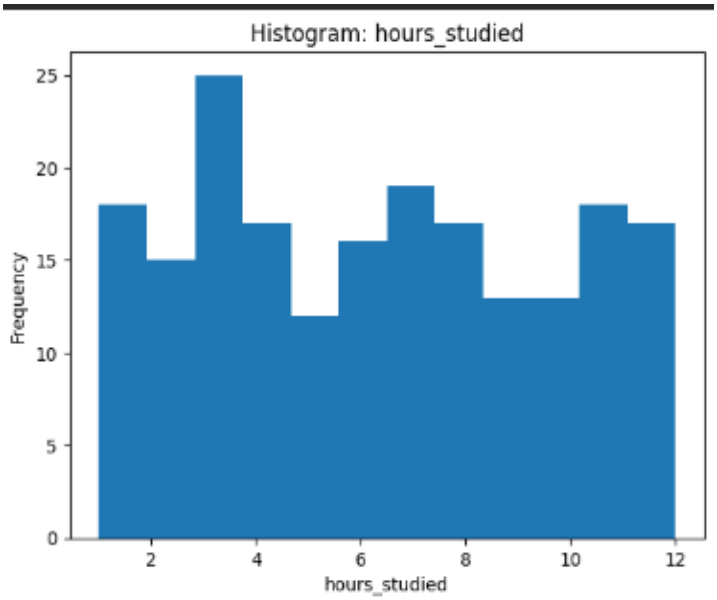
Continuous attributes were discretized into three categories (Low, Medium, High) using equal-width binning:

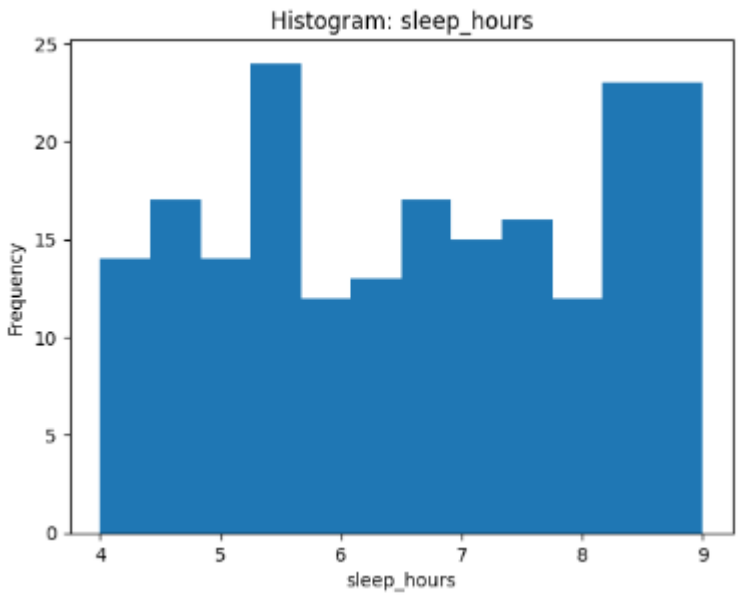
- hours_studied: Binned into Low, Medium, High based on study hour ranges
- sleep_hours: Categorized into sleep quality levels
- previous_scores and exam_score: Grouped into performance levels

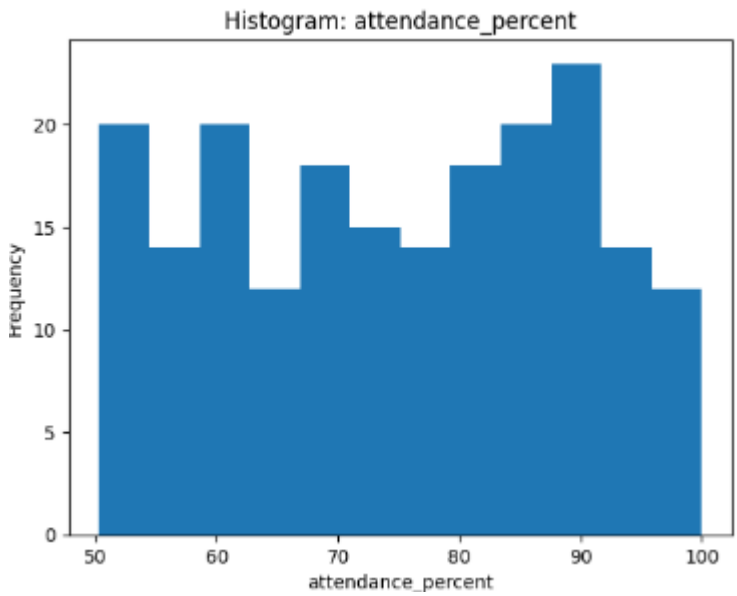
Visualization

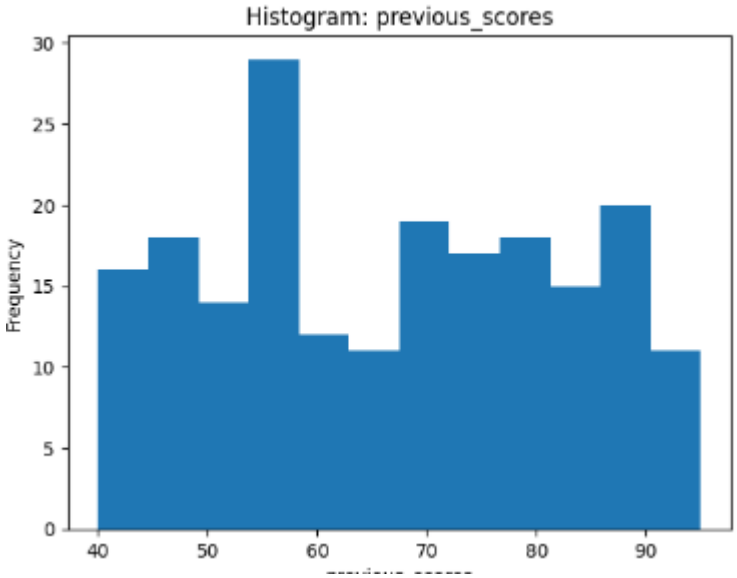
Exploratory data analysis was conducted using:

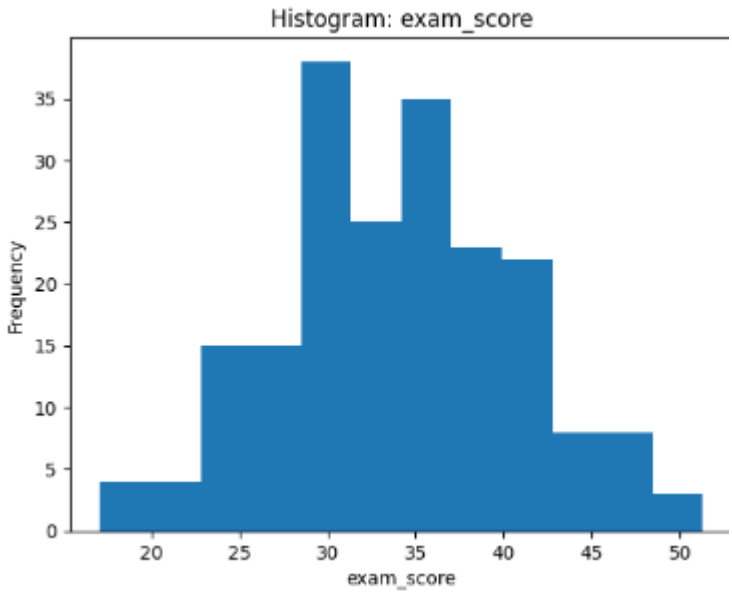
- **Histograms for categorical attributes**




Histogram: sleep_hours
Frequency
sleep_hours
0
5
10
15
20
25
4
5
6
7
8
9


Histogram: attendance_percent
Frequency
attendance_percent
0
5
10
15
20
50
60
70
80
90
100





Conclusion:

Histograms provided an overview of the distribution of each numerical feature:-

- Hours Studied showed a right-skewed pattern — most students study 4–6 hours daily, while a few study above 10 hours.
- Sleep Hours centered around 7 hours, aligning with healthy sleep behavior.
- Attendance Percent was concentrated between 75–95%, suggesting generally consistent attendance among students

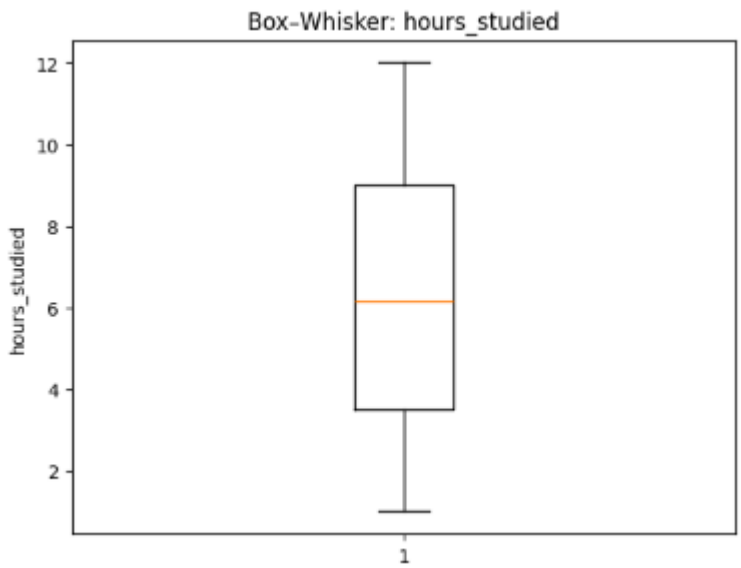
- **Distribution plots for continuous attributes**

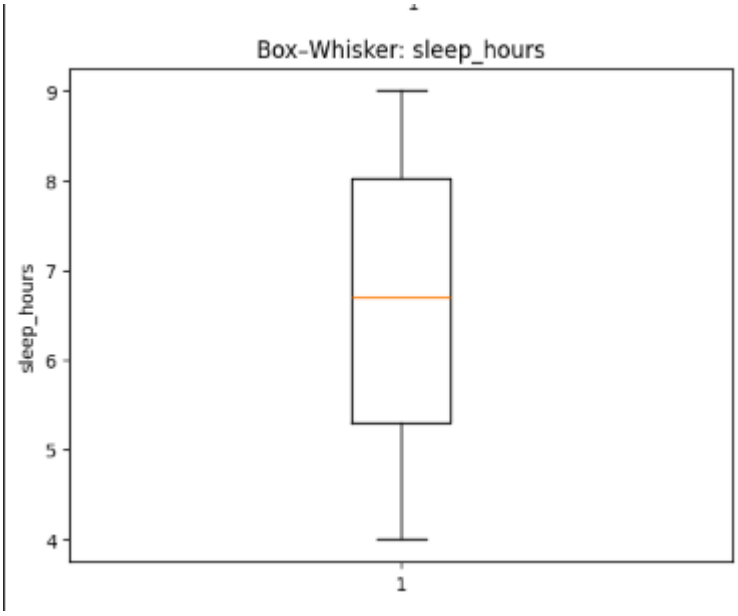
	mean	median	std	skew	kurtosis
hours_studied	6.326	6.15	3.227	0.117	-1.218
sleep_hours	6.622	6.70	1.497	-0.043	-1.295
attendance_percent	74.830	75.25	14.250	-0.073	-1.198
previous_scores	66.800	67.50	15.664	0.060	-1.220
exam_score	33.955	34.05	6.790	0.033	-0.346

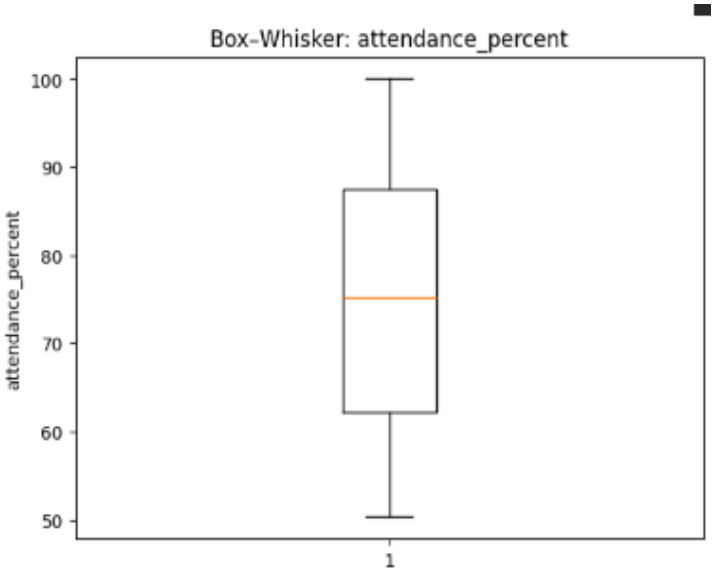
<u>Conclusion:</u>

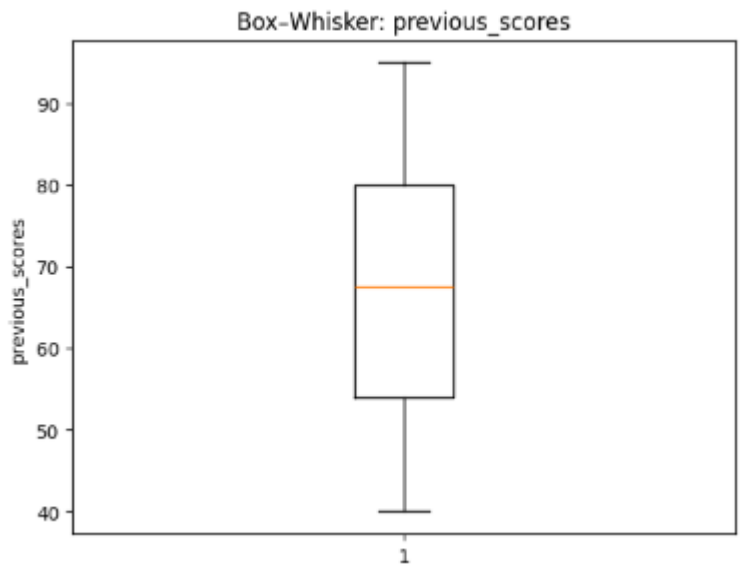
The distributions are mostly balanced with slight skewness but nothing extreme. Study hours and previous scores show moderate spread, while sleep hours vary less. Attendance is generally high with some variation. Overall, the data looks normal with no major outliers or heavy tails.

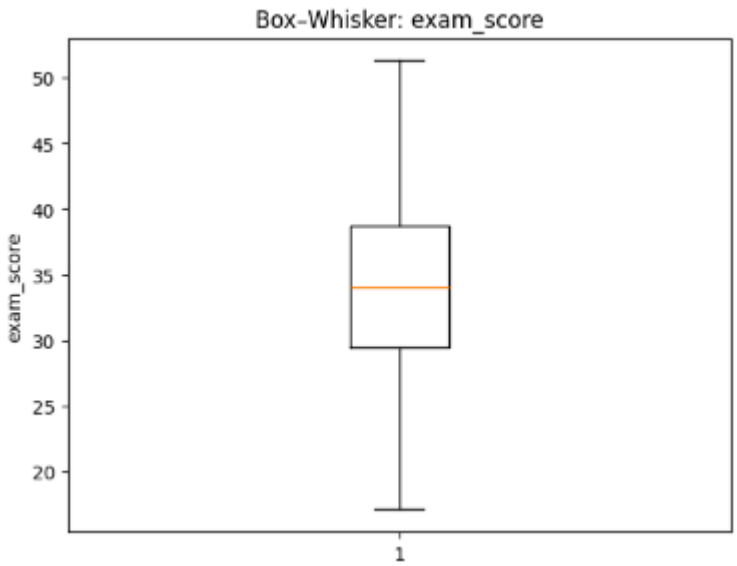
- **Box Plot to identify outliers**




Box-Whisker: sleep_hours
sleep_hours
9
8
7
6
5
4
1


Box-Whisker: attendance_percent
attendance_percent
100
90
80
70
60
50
1



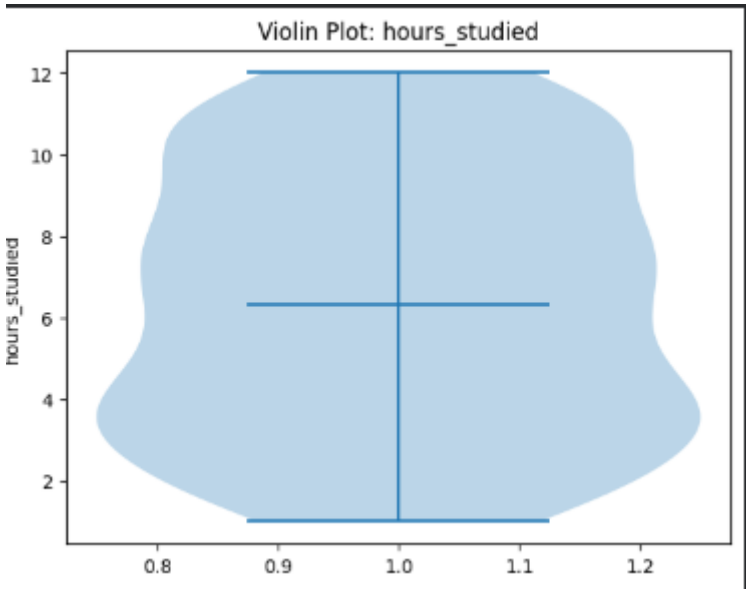


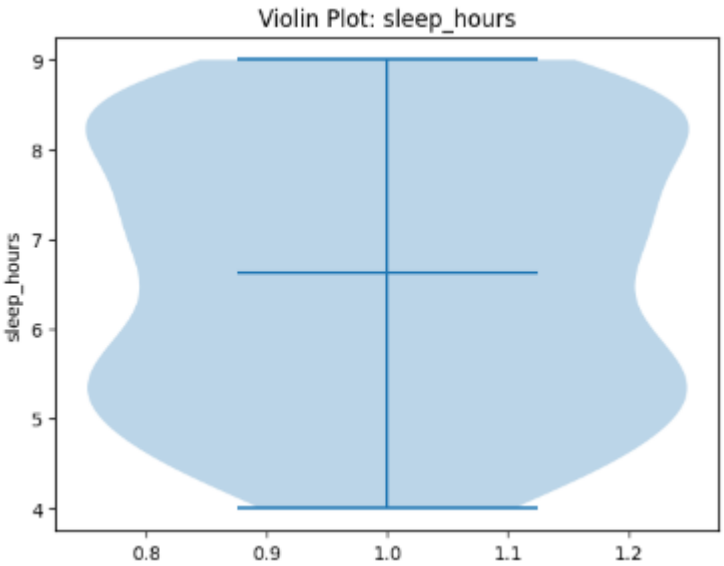
Conclusion

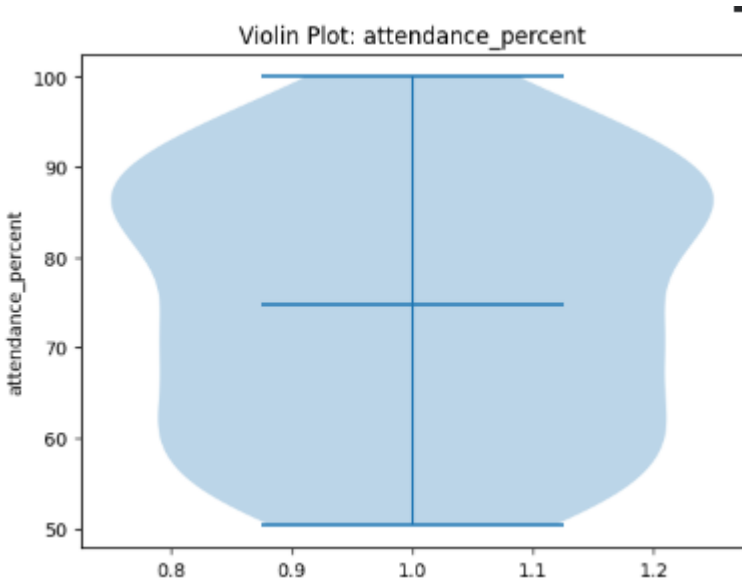
Box–Whisker plots revealed:-

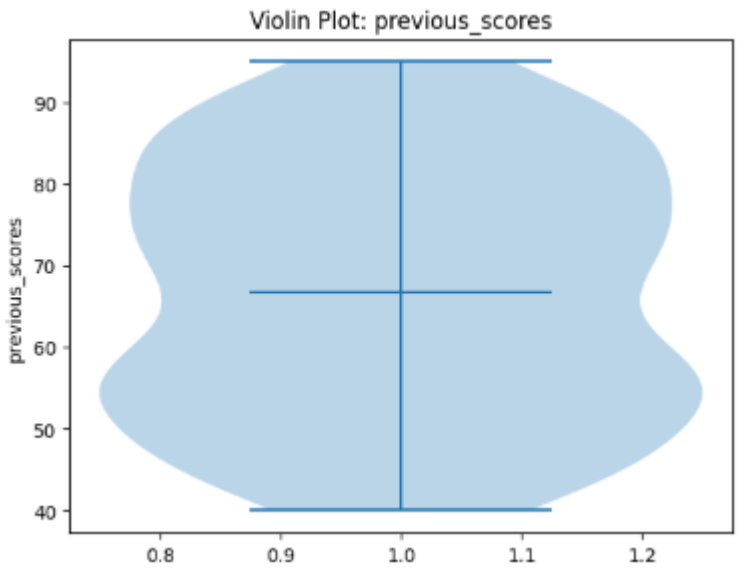
- A few outliers in hours_studied (very high study hours) and sleep_hours (low sleep duration).
- attendance_percent was tightly packed with minimal outliers, suggesting consistent attendance patterns.
- Previous_scores and exam_score showed mild spread but no severe anomalies

- **Whisker plots to identify outliers**


Violin Plot: hours_studied
hours_studied
12
10
8
6
4
2
0.8
0.9
1.0
1.1
1.2


Violin Plot: sleep_hours
sleep_hours
9
8
7
6
5
4
0.8
0.9
1.0
1.1
1.2


Violin Plot: attendance_percent
attendance_percent
100
90
80
70
60
50
0.8
0.9
1.0
1.1
1.2

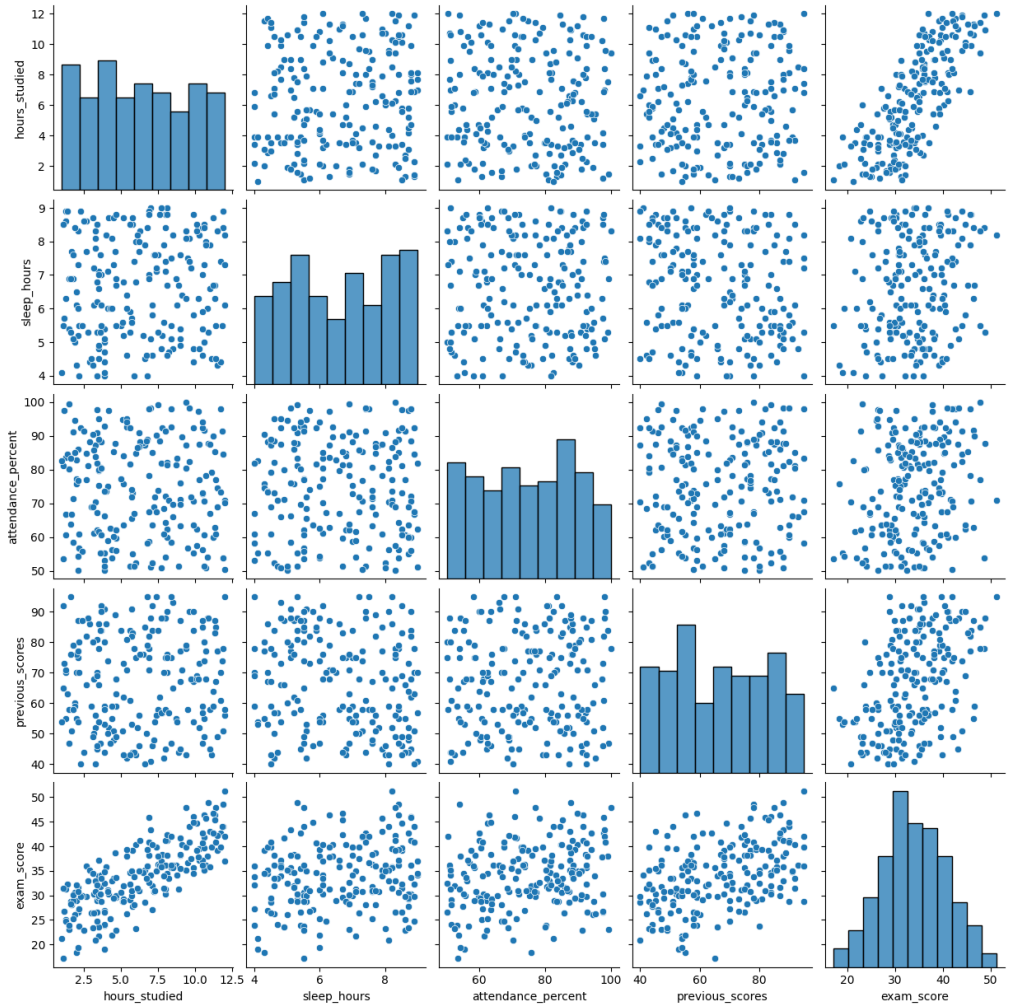


Conclusion:

Violin plots helped visualize the density and spread of each variable:-

- Hours Studied had a thick base near 4–6, confirming the dominant group of moderate learners.
- Exam Score density hinted at distinct clusters, supporting the future use of clustering to group students by performance.
- Sleep Hours displayed a symmetric distribution, while Previous Scores leaned toward higher values.

- **Scatter plot matrices to examine relationships between variables**



Conclusion:

Overall, the scatterplot shows that hours studied has the strongest positive relationship with exam score .Students who study more tend to score higher. The other factors (sleep hours, attendance, and previous scores) show little to no clear correlation with exam performance in this dataset. This suggests that, among all variables measured, studying hours are the most reliable predictor of exam success.

Dimensionality Reduction

PCA

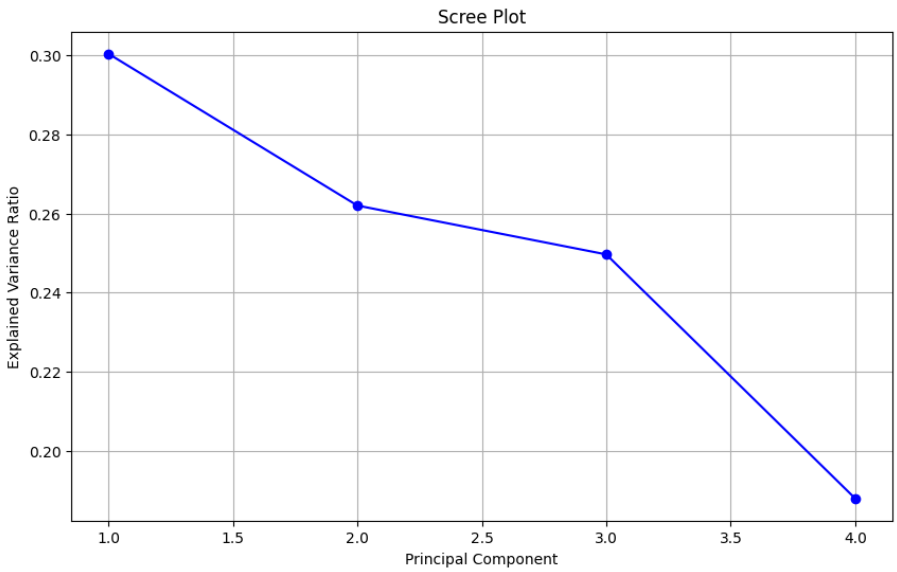


Figure 4 : Scree Plot

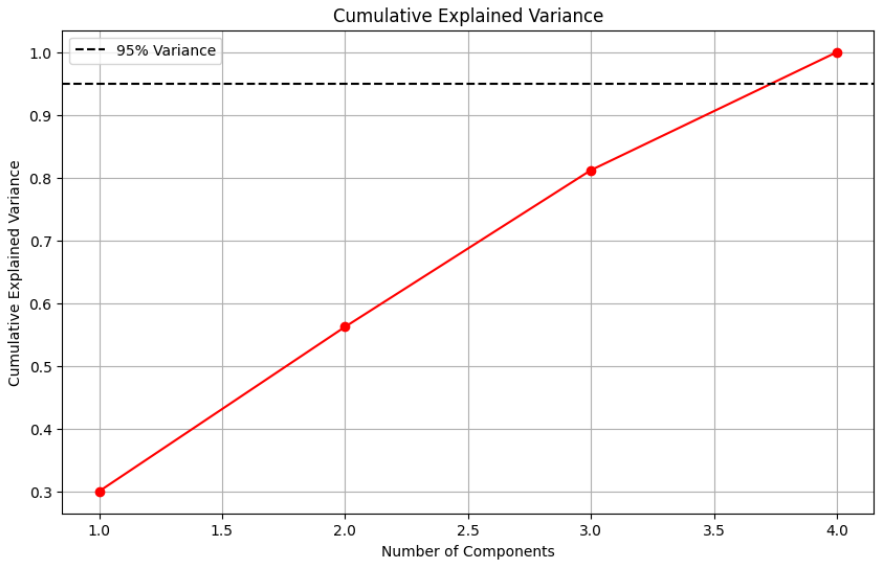


Figure 5: Cumulative Explained Variance Plot

<u>Conclusion:</u>

The PCA results show that the dataset's variability is well captured by the first few principal components. The scree plot indicates that PC1, PC2, and PC3 explain the largest share of

variance, with decreasing contributions from each subsequent component. The variance ratios explained confirm this pattern, with the first three components accounting for approximately 81% of the total variance, while the fourth component contributes the remaining 19%.

The cumulative explained variance plot further demonstrates that three components provide a strong lower-dimensional representation, though all four components are required to reach nearly 100% variance, exceeding the 95% threshold typically used for full information retention. The PCA-transformed data shows clear numerical separation along these components, suggesting that PCA effectively captures meaningful underlying structure in the dataset. Overall, the analysis indicates that reducing the data to 2–3 principal components is sufficient for visualization and clustering, while including all four components preserves the complete variance of the original dataset.

Feature Selection and Generation

Selected Features

Five continuous attributes were selected for clustering analysis:

- hours_studied
- sleep_hours
- attendance_percent
- previous_scores
- exam_score

```
Selected features: ['hours_studied', 'sleep_hours', 'attendance_percent', 'previous_scores', 'exam_score']
Shape: (200, 5)
```

Figure 6 : Selected Features

Feature Scaling

StandardScaler was applied to normalize all features to have zero meaning and unit variance. This ensures that no single feature dominates the distance calculations in clustering algorithms due to differences in scale.

```
Features standardized
   hours_studied  sleep_hours  attendance_percent  previous_scores  exam_score
0       0.520154     1.458426           -0.192061        -1.395230   -0.554444
1      -1.561083     1.324503           -0.994074        -0.755216   -1.322249
2      -0.722376     1.056656           -0.079498         1.228827    0.272423
3      -0.877692    -1.220042            1.426035        -0.051201    0.006644
4       0.861850    -0.148655            1.053170         0.268806    0.936870
```

Figure 7: Features standardized

Clustering Methods

K-means Clustering

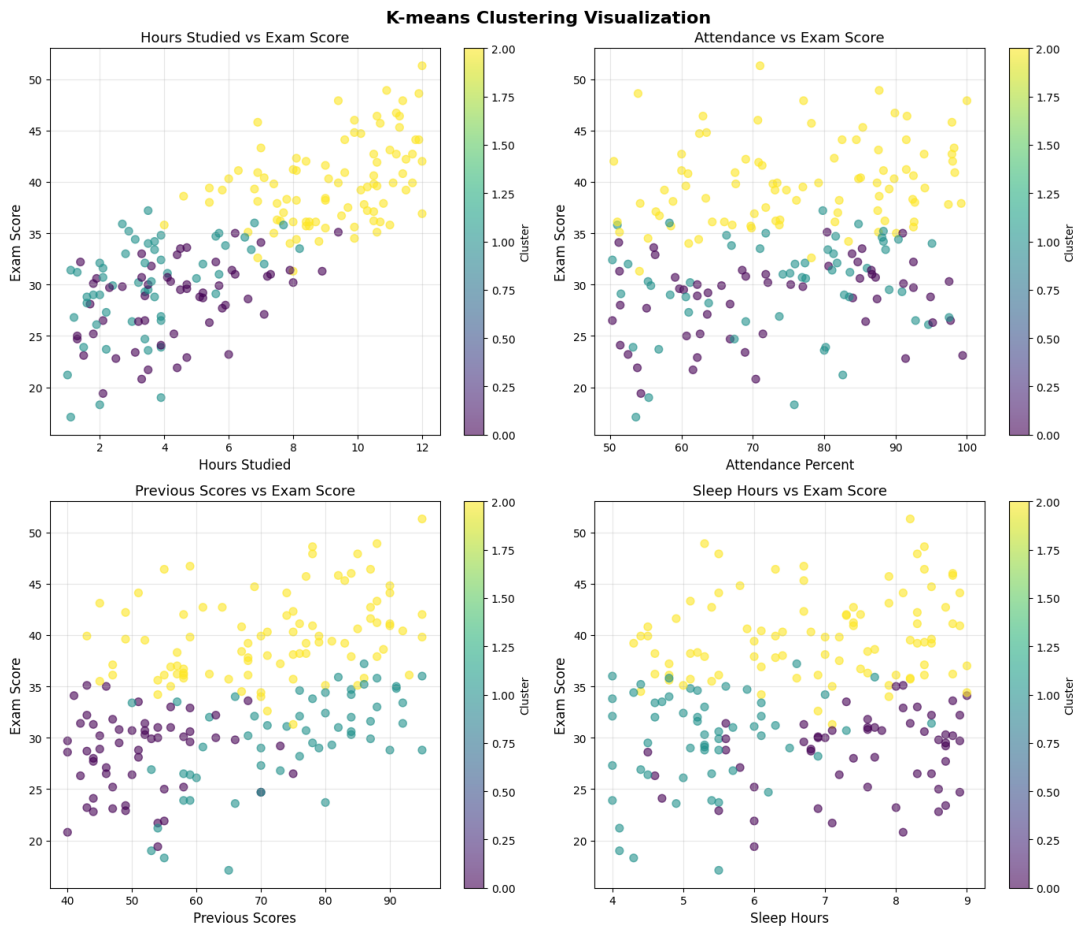


Figure 7: K-means Clustering

K-means clustering was applied with k=3 clusters. The algorithm partitions the data by:

- Randomly initializing 3 cluster centroids
- Assigning each student to the nearest centroid based on Euclidean distance
- Iteratively updating centroids until convergence

The algorithm successfully converged and produced three distinct student groups with relatively balanced cluster sizes.

Hierarchical Clustering

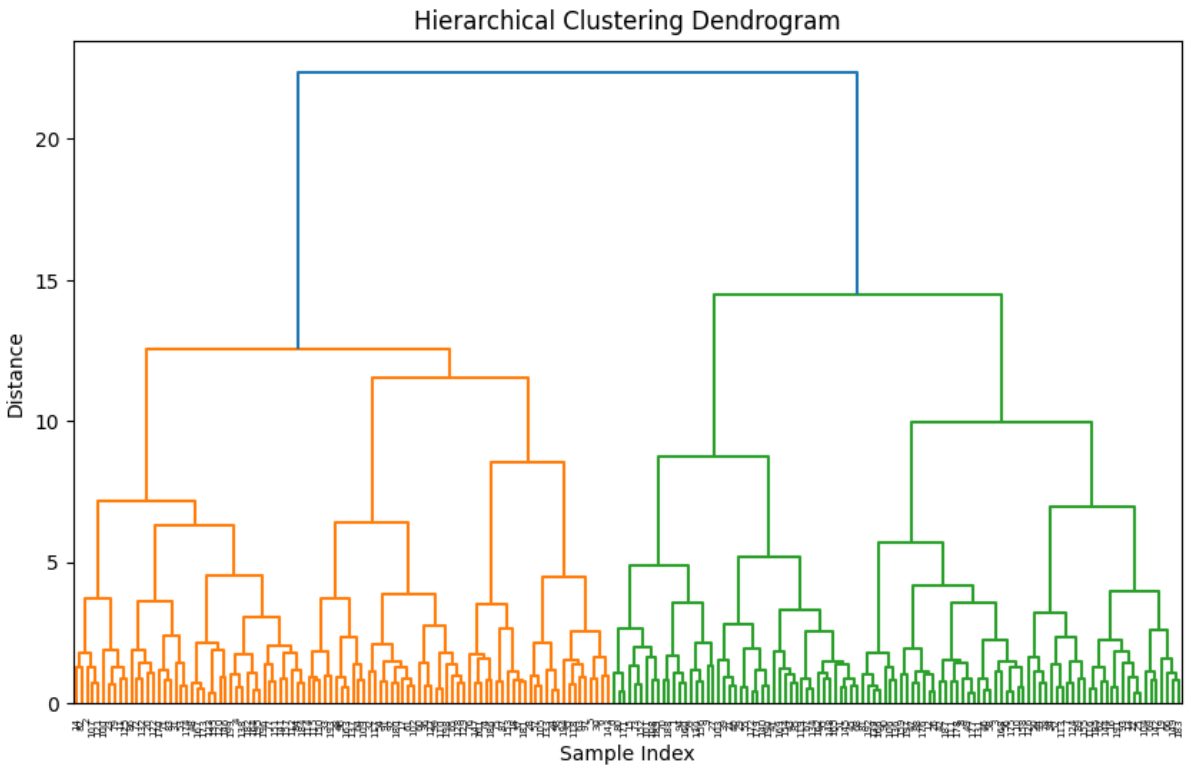


Figure 8: Hierarchical Clustering

Agglomerative hierarchical clustering was performed using Ward's linkage method. This approach:

- Starts with each student as a separate cluster
- Merges the two closest clusters at each step based on minimum variance increase

- Creates a dendrogram showing the hierarchical relationship between clusters

The dendrogram was cut to produce 3 clusters in comparison with K-means results.

Cluster Evaluation

Both clustering methods were evaluated using multiple internal validation metrics:

Silhouette Score

- K-means (0.2209) is higher than hierarchical (0.1850).
- This indicates that K-means forms slightly better-defined and more separated clusters, with points fitting their assigned clusters more appropriately.

Davies-Bouldin Index

- K-means (1.4524) is lower than hierarchical (1.4885).
- Since lower is better, K-means shows tighter clusters with less overlap, meaning its clusters are more compact and distinct.

Calinski-Harabasz Index

- K-means (64.8582) is higher than hierarchical (54.0493).
- This means K-means achieves better between-cluster separation and lower within-cluster variance, indicating stronger overall cluster structure.

Evaluation Results

The clustering algorithms produced the following evaluation metrics:

Metric	K-means	Hierarchical
Silhouette Score	0.2209	0.1850
Davies-Bouldin	1.4524	1.4885
Calinski-Harabasz	64.6562	54.0493

Results and Discussion

Cluster Characteristics

The clustering analysis identified three distinct student groups:

Cluster 1: High Performers

- Students with high study hours and strong attendance
- Consistently high exam scores
- Balanced sleep patterns

Cluster 2: Average Performers

- Moderate study hours and attendance
- Average exam performance
- Variable sleep patterns

Cluster 3: At-Risk Students

- Low study hours and irregular attendance
- Below-average exam scores

- Inconsistent sleep patterns

Algorithm Comparison

Both K-means and hierarchical clustering produced meaningful groupings, though with some differences:

- K-means produced more evenly sized clusters
- Hierarchical clustering revealed the nested structure of student groups
- Both methods identified similar high-performing and at-risk student groups

Final Conclusions

This clustering analysis successfully identified distinct student groups based on study habits, attendance, sleep patterns, and academic performance. The key findings are:

- Three distinct student profiles emerged from the data
- Study hours and attendance are strong indicators of cluster membership
- Both clustering methods produced comparable and meaningful results
- The identified clusters can inform targeted educational interventions

Practical Applications

The clustering results can be used to:

- Identify at-risk students early in the semester
- Design targeted support programs for different student groups
- Allocate educational resources more effectively
- Develop personalized learning strategies

Future Work

Potential extensions of this work include:

- Testing different numbers of clusters using elbow method or silhouette analysis
- Incorporating additional features such as socioeconomic factors
- Applying other clustering algorithms such as DBSCAN or Gaussian Mixture Models
- Validating clusters against external criteria such as graduation rates

References

- Scikit-learn: Machine Learning in Python. Pedregosa et al., JMLR 12, pp. 2825-2830, 2011
- Python Data Science Handbook by Jake VanderPlas
- Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar. Introduction to Data Mining, 2nd Ed., Morgan Kaufmann, 2011